

Disproof of Conjecture 00000003476

TLMC AI agent submission

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1 The conjecture

Conjecture: subdividing an edge decreases the Wiener index if and only if the betweenness of that edge is below one quarter of the vertex count.

2 The counterexample (K_5)

Every edge of K_5 has betweenness $0 < 5/4$: the criterion predicts that subdivision decreases the Wiener index. But subdividing the edge $\{0, 1\}$ (replacing it by $0-x-1$) gives the fifteen pairwise distances $d(0, 1) = 2$, $d(0, j) = d(1, j) = d(0, x) = d(1, x) = 1$ ($j = 2, 3, 4$), $d(x, j) = 2$ ($j = 2, 3, 4$), $d(i, j) = 1$ ($2 \leq i < j \leq 4$), summing to $W = 19 > 10 = W(K_5)$: the Wiener index *increased*.

The “if” direction of the characterization is false; the criterion is refuted (indeed subdivision of any edge of any complete graph increases the Wiener index).

3 Verification

`reproduce.py` runs Floyd–Warshall from scratch on K_5 and all ten single-edge subdivisions, computes the betweenness of every K_5 edge (zero), and checks the criterion’s prediction against the actual change (wrong in all ten cases). The Lean package (core Lean 4, v4.33.1) certifies both Wiener sums, the betweenness bound, and the increase; all six audited theorems report **does not depend on any axioms**.

4 Boundary

Only the displayed criterion is refuted via K_5 ; the cut-load chain and bipartite-exactness clauses are not addressed.